

- 4 A device to measure the alternating currents in cables consists of a toroidal solenoid wound on a plastic ring. The current-carrying cable is placed along the axis through the centre of the device as shown in Fig. 4.1. The radius of the plastic ring r is 0.150 m. The toroidal solenoid has 6000 turns and each turn has a radius of 0.010 m.

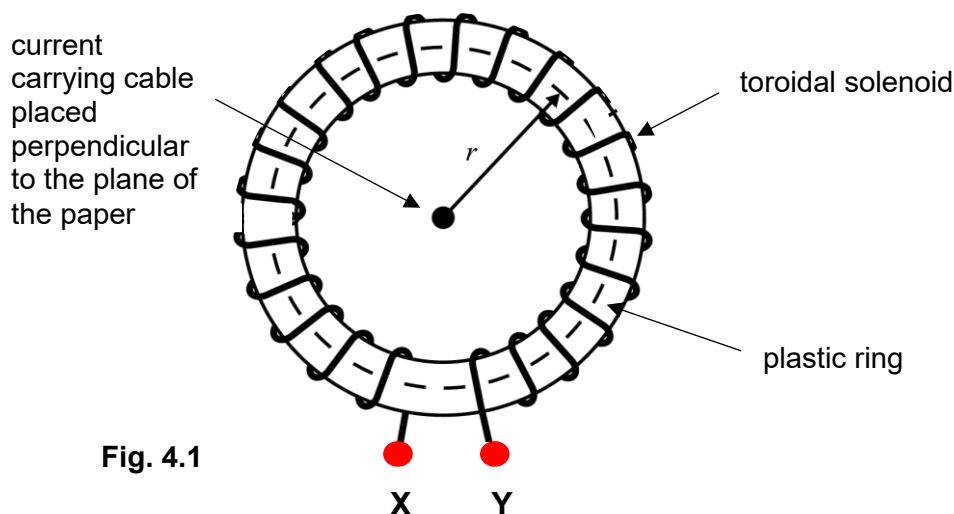


Fig. 4.1

The sinusoidal current I flowing in the cable is as shown in Fig. 4.2.

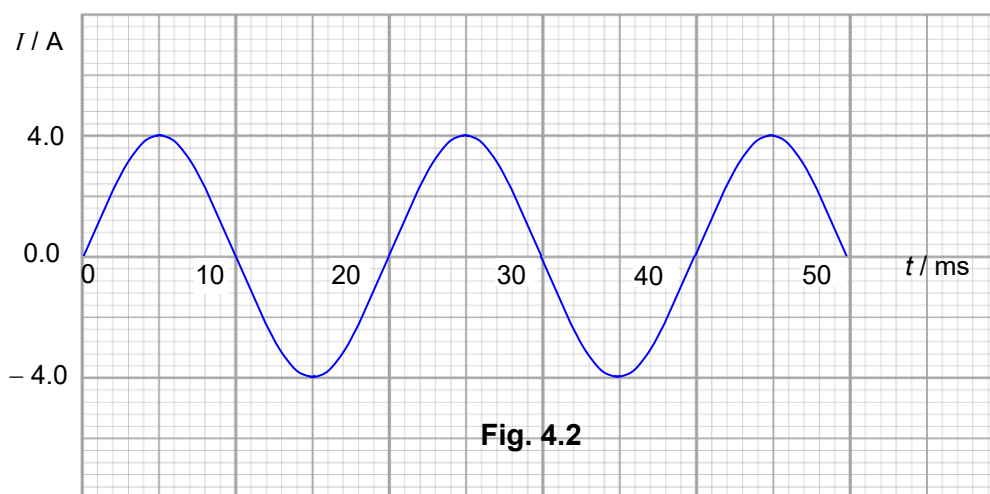


Fig. 4.2

- (a) Show that the expression for the magnetic flux density B at the centre of the toroidal solenoid due to the current carrying cable is $B = 5.33 \times 10^{-6} \sin 314 t$.

(b) The terminals XY is connected to an oscilloscope.

- (i) Using the laws of electromagnetic induction, explain why a voltage is recorded by the oscilloscope.

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 [2]

- (ii) Determine the peak induced e.m.f. detected by the oscilloscope.

peak induced e.m.f =V [3]

- (iii) A diode is added as shown in the Fig. 4.3.

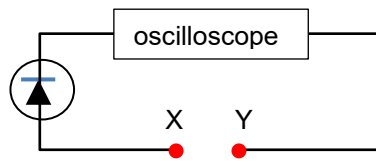


Fig. 4.3

On Fig 4.4, sketch the waveform of the voltage displaced on the oscilloscope over two cycles.



Fig. 4.4

[2]